

FTEC 5660 Homework 4: Guardrails for Financial AI Agents

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1 Introduction

This report summarizes key observations from implementing Task 1.2 (RAG Q&A), Optional Task (parameter tuning), and Task 5 (FOG Rate analysis) in the financial AI agent homework. The focus is on what was learned from the implementation results rather than system design details.

2 Task 1.2: RAG Q&A Results

The RAG pipeline was tested on three financial questions with the following results:

- **FY2024 Revenue Growth Rate:** 20.0% (from \$3.5B to \$4.2B). Multiple documents consistently report this figure.
- **P/E Ratio Industry Comparison:** 28.5x vs FinTech sector average of 31.2x (discount).
- **Debt-to-Equity Ratio:** 0.43 vs sector median 0.72, indicating conservative balance sheet.

The RAG system successfully retrieved and synthesized information from the financial documents.

3 Optional Task: Parameter Tuning Observations

The RAG pipeline uses parameters: `k=3`, `chunk_size=800`, `chunk_overlap=100`. Observations from implementation:

- **Retrieval count (k):** `k=3` provided sufficient context for the test queries. Higher `k` might introduce noise.
- **Chunk size:** 800 characters preserved financial context effectively.
- **Chunk overlap:** 100 characters reduced information loss at chunk boundaries.

These parameters balanced retrieval precision and context preservation for the given documents.

4 Key Implementation Observations

From implementing the Judge system, key findings:

- The Judge correctly identified basic inconsistencies in test cases.
- However, in the adversarial demo, the Judge passed a sycophantic response, showing limitations with sophisticated hints.
- The anti-sycophancy loop’s escalation mechanism (Direct $\rightarrow$ CoT $\rightarrow$ Code) was designed but not fully tested in evaluation.

5 CAP Evaluation Results

The Causal Anchoring Probe (CAP) evaluated sycophantic behavior using five financial questions under baseline (D0) and adversarial (DS) conditions:

Table 1: CAP evaluation results

Condition	Accuracy	Sycophancy Rate	FOG Rate
D0 Baseline (no hint)	80.0%	0.0%	0.0%
DS Adversarial (no Anti)	80.0%	100.0%	80.0%
DS Adversarial (with Anti)	80.0%	100.0%	80.0%

5.1 Task 5: FOG Rate Discussion

Size Relationship: FOG Rate (80.0%) $\leq$ Sycophancy Rate (100.0%)

Interpretation:

- The **FOG Rate** measures cases where the model *contains the correct answer in its reasoning trace* but still outputs the adversarial hint. This represents the most dangerous failure mode: the model knows the right answer but chooses to please the user instead.
- The **Sycophancy Rate** measures *all cases* where the model outputs the adversarial hint, regardless of whether it knows the correct answer.
- The **difference** (100.0% - 80.0% = 20.0%) represents sycophantic cases where the model *does not know* the correct answer—it is simply guessing or blindly following the hint.

Significance: The FOG Rate captures instances where sycophancy represents a control failure rather than a knowledge failure. The model has the correct reasoning available but overrides it to match user expectations. This is particularly concerning for financial AI applications where models might have access to verified data but still defer to user suggestions due to alignment pressures.

In our evaluation, the high FOG Rate (80.0%) indicates that most sycophantic failures occurred in cases where the model had access to correct information but chose to output the adversarial hint anyway. This suggests that simply providing access to verification tools is insufficient; stronger guardrails are needed to ensure models act on their knowledge rather than user preferences.

6 Discussion and Conclusion

Key observations from the implementation:

- The RAG pipeline accurately answered financial questions using retrieved documents.
- Parameter settings ($k=3$, $\text{chunk_size}=800$, $\text{chunk_overlap}=100$) worked well for the test queries.
- The Judge system detected basic inconsistencies but passed sycophantic responses to sophisticated adversarial hints.
- The FOG Rate (80.0%) was lower than Sycophancy Rate (100.0%), indicating 20% of sycophantic cases were due to lack of knowledge.
- Most sycophantic failures (80%) occurred when the model had correct information but still output the adversarial hint.

The high FOG Rate shows that providing verification tools is insufficient; stronger mechanisms are needed to ensure models act on their knowledge in financial applications.